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ALL THE BEST FOR YOUR EXAM

Discrete Mathematics — Complete Question Bank (Unit Wise)
UNIT I — Set and Propositions

Propositions & Logic

1. Let p be "Mark is Rich" and q be "Mark is happy". Write each of the following in symbolic form:

- i) Mark is poor but happy
- ii) Mark is neither rich nor happy
- iii) Mark is either rich or happy
- iv) Mark is Rich and not happy

2. Explain terms Tautology and Contradiction in truth table with an example.

3. Let p and q be the propositions — p : It is below freezing. q : It is snowing. Write these propositions using p and q and logical connectives:

- i) It is below freezing and snowing.
- ii) It is below freezing but not snowing.
- iii) It is either snowing or below freezing (or both).
- iv) If it is below freezing, it is also snowing.
- v) Either it is below freezing or it is snowing, but it is not snowing if it is below freezing.

4. Let P : I will study hard and Q : I will get admission in IIT. Statement: If I study hard then I will get admission in IIT. Write the Converse, Inverse & Contrapositive of the above statement.

5. Express the contrapositive, converse and inverse form of the conditional statement: "If x is rational, then x is real".

6. What is Logical Equivalence? Show that $\sim(q \rightarrow p) \leftrightarrow (p \rightarrow q) \rightarrow q$

Mathematical Induction

7. By using mathematical induction show that: $1 + 2 + 3 + \dots + n = n(n+1)/2$ for all natural number values of n .

8. By using mathematical induction prove that: $S_n = 1 + 3 + \dots + (2n-1) = n^2$; for all integers $n > 1$

9. Use mathematical induction to prove: $S_n = 2 + 4 + 6 + 8 + \dots + 2n = n(n+1)$ for all positive integer n .

10. By using mathematical induction show that: $1 + 4 + 7 + \dots + (3n-2) = n(3n-1)/2$ for all natural number values of n .

Sets & Set Operations

11. Let $U = \{1,2,3,\dots,10\}$, $A = \{2,4,6,8,10\}$, $B = \{1,3,5,7,9,10\}$. Find:

- i) $(A \cup B)'$
- ii) $(A \cap B)'$
- iii) $(B)'$
- iv) $(B-A)'$

12. Let $A = \{1,2,3\}$ and $B = \{1,2,3,4,5\}$. Find:

- i) $P(A \cup B)$
- ii) $P(A \cap B)$
- iii) $A - B$

13. Solve the following:

- i) Find the cardinality of Power Set of A , where $A = \{1,2,9\}$

- ii) If $A = \{1,2\}$, $B = \{2,3,4\}$, $C = \{4,5\}$, find: $A \times (B \cap C)$
- iii) Let $A = \{1,2,3\}$, $B = \{1,2,3,4,5\}$. Find: $P(A \cap B)$, $P(A \cup B)$, $A - B$

14. Explain the following terms with example:

- i) Symmetric difference between sets
- ii) Union of set
- iii) Intersection of Set
- iv) Subset of a Set

15. Explain the following terms with example:

- i) Symmetric difference between set
- ii) Universal Set
- iii) Complement of a Set
- iv) Power Set
- v) Proper subset

Inclusion-Exclusion Principle

16. A college records gives following information: 119 students enrolled in Introductory Computer Science, 96 took Data Structures, 53 took Foundations, 39 took Assembly Language, 31 took both Foundation and Assembly Language, 32 took both Data Structures and Assembly Language, 38 took Data Structures and Foundations, and 22 took all three courses. Is this information correct? Why?

17. Suppose 100 Computer Engineering students study at least one of C, C++, and Python. 65 study C, 45 study C++, 42 study Python. 20 study C and C++, 25 study C and Python, 15 study C++ and Python. Find students studying:

- i) Only C and C++, not Python
- ii) Only C and Python, not C++

18. A software company employs 100 programmers. 45 are proficient in Java, 30 in C#, 20 in Python, 6 in C# and Java, 1 in Java and Python, 5 in C# and Python, 1 in all three. Determine programmers not proficient in any of these three languages.

19. In a survey of 60 people: 25 read Newsweek, 26 read Time, 26 read Fortune, 9 read Newsweek and Fortune, 11 read Newsweek and Time, 8 read Time and Fortune, 8 read no magazine.

- i) Find total people who read all three magazines.
 - ii) Fill correct numbers in Venn diagram.
 - iii) Determine people who read exactly one magazine.
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UNIT II — Relations and Functions

Relations

20. What is Equivalence Relation? Explain properties of binary relations.
21. Let $A = \{1,2,3,4\}$ and $R = \{(1,2),(2,4),(1,3),(3,2)\}$. Find transitive closure of relation R using Warshall's algorithm.
22. Let $A = \{1,2,3,4,12\} = B$, and let aRb if a divides b . Write the relation and draw its Hasse diagram.
23. Let $A = \{0,2,4,6,8,10\}$ and relation $aRb = \{(a,b) \mid (a-b) \% 2 == 0; a,b \in A\}$. Find if aRb is an Equivalence Relation or not.
24. Write the relation pairs and draw the Hasse Diagram for $aRb = \{(a,b) \mid a \text{ divides } b; a,b \in X\}$, where $X = \{10,20,30,40,50,60,80,100\}$.
25. If $X = \{10,20,30,40,50\}$ and $aRb = \{(a,b) \mid a \text{ divides } b; a,b \in X\}$. Find whether aRb is a Partial Order Relation or not.
26. Let $A = \{1,2,4,8,16,24,32,48\}$. Relation $aRb = \{(a,b) \mid a \text{ divides } b; a,b \in A\}$.
- i) Write Relation aRb
 - ii) Write any two Chains of aRb on set 'A'
 - iii) Write any two Anti Chains of aRb on set 'A'
27. What is Equivalence relation? Let $A = \{1,2,3,4,5\}$ and $R = \{(1,1),(1,2),(2,1),(1,3),(1,4),(4,5),(5,1),(1,5),(4,1)\}$. Draw digraph of R .
28. Let R be a relation on Set $A = \{1,2,3,4\}$, given as $R = \{(1,1),(1,4),(2,2),(2,3),(3,2),(3,3),(4,1),(4,4)\}$. Find transitive closure using Warshall's Algorithm.
29. Construct the Hasse diagram for "a divides b" relation on the set $\{2,3,4,6,8,9,12,18\}$.
30. For each of these relations on Set $A = \{1,2,3,4\}$, decide whether it is reflexive, symmetric, transitive or anti-symmetric:
- $R_1 = \{(1,1),(2,2),(3,3),(4,4)\}$
 - $R_2 = \{(1,1),(1,2),(2,2),(2,1),(3,3),(4,4)\}$
 - $R_3 = \{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$
31. Let $A = \{1,2,3,4\}$ and $R = \{(1,2),(2,1),(2,3),(3,4)\}$. Find transitive closure of relation R using Warshall's algorithm.
32. Draw a Hasse diagram for $(S, \leq)$ where $S = \{1,2,3,4,5,6\}$ and $\leq$ is defined as $a \leq b$ if a divides b .
33. Find whether the following posets are lattices or not. (Diagram in Oct 2022 and Oct/Nov 2025 papers)

Functions

34. Let $f(x) = 2x+3$, $g(x) = 3x+4$, $h(x) = 4x$. Find gof , fog , foh , goh .
35. If $f(x) = 2x+5$ and $g(x) = 5x+2$, find:
- i) $\text{fog}(5)$

- ii) $\text{fog}(2) + \text{gof}(2)$

36. If $f(x) = 16x^2 + 12$. Find Inverse of $f(x)$. Is the inverse of $f(x)$ a function? Justify.

37. Let $f(x) = x+2$, $g(x) = x-2$, $h(x) = 3x$. Find gof , fog , fof , gog , foh .

38. Explain the various types of functions.

39. Using the function f and g , check whether $\text{fog} = \text{gof}$, where $f(x) = 4x^2 - 1$, $g(x) = 1+x$.

40. If $f(x) = x^2 - 1$, $g(x) = x - 2$, find a if $\text{gof}(a) = 1$. Also find k , if $f(k) = 2k - 1$ and $\text{fof}(k) = 5$.

ABHYASMITRA

UNIT III — Introduction to Trees

Tree Definitions & Properties

41. Define the following terms with suitable example:

- i) Forest
- ii) Fundamental cutsets
- iii) Game tree

42. Define the following terms:

- i) Level of a tree
- ii) Height of a tree
- iii) Fundamental circuit

43. Define the following terms with example:

- i) Binary tree
- ii) Ordered tree
- iii) Eccentricity of vertex

44. Define with example:

- i) Level and height of a tree
- ii) Binary search tree
- iii) Spanning tree

45. Explain:

- i) Cutset
- ii) Tree properties
- iii) Prefix code

46. Explain:

- i) Cut set
- ii) Complete Binary tree
- iii) Prefix code

47. Construct a binary tree from given inorder and preorder traversals:

- Inorder: b d f h k m p t v w
- Preorder: b f d k h v w t m

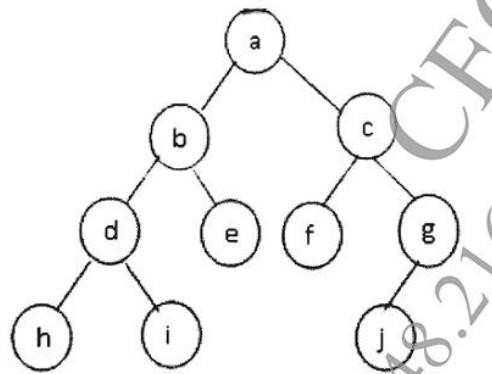
48. Construct binary search tree by inserting integers in order: 50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. Find:

- i) No. of internal nodes
- ii) Leaf nodes

49. Construct Binary Search Tree with: 21, 28, 14, 18, 11, 32, 25, 23, 37, 27, 5, 15, 19, 30, 12, 26.

. Explain the following:

- i) Regular m-ary tree with example
- ii) Find the preorder, postorder and inorder of the given binary tree



51. What is Minimum Spanning Tree? Explain briefly the steps involved in finding MST using Prim's Algorithm.

Huffman Coding / Optimal Trees

52. Construct an optimal binary tree for the set of weights {8,9,10,11,13,15,22}. Find the weight of an optimal tree. Also assign the prefix codes and write the code words.

53. Construct a Huffman tree for:

Symbol Frequency

A	5
B	6
C	6
D	11
E	20

54. Construct an optimal tree for 10, 11, 14, 21, 16, 18 using Huffman coding.

55. Find the optimal prefix code for the given characters with frequency of occurrences:

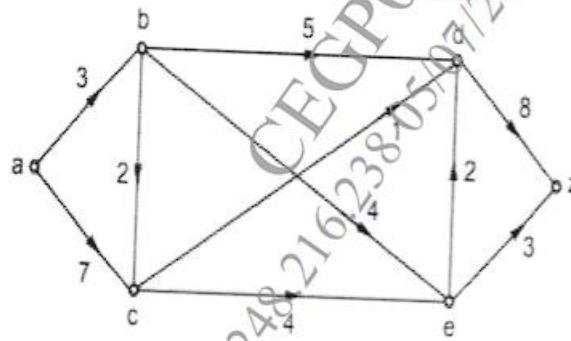
Character Frequency

A	10
E	15
I	12
O	3
U	4
S	13
T	1

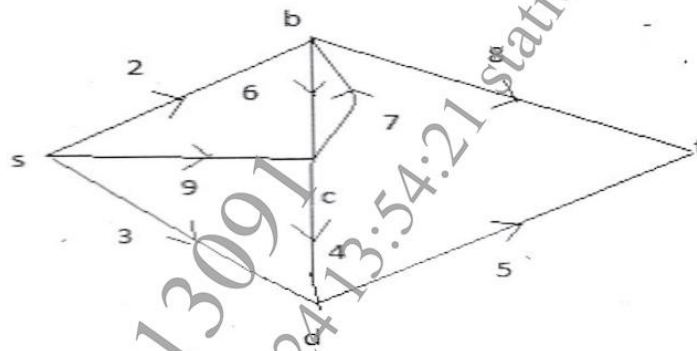
56. Suppose data items A, B, C, D, E, F, G occur in frequencies 8, 22, 9, 11, 13, 10, 15 respectively. Construct Huffman code for the data.

Maximum Flow — Min Cut

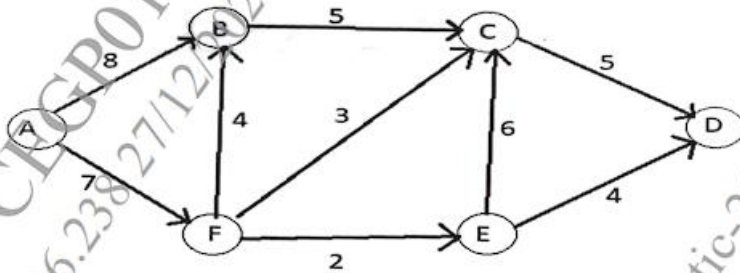
57. Find maximum flow in the transport network using labeling procedure. Determine the corresponding min-cut.



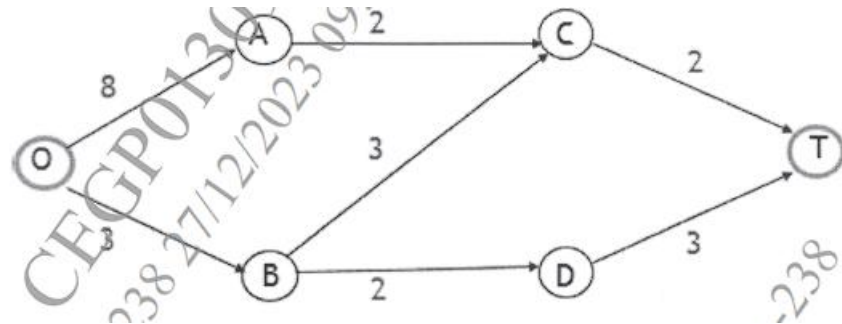
58. Use labeling procedure to find a maximum flow in the transport network. Determine the corresponding minimum cut.



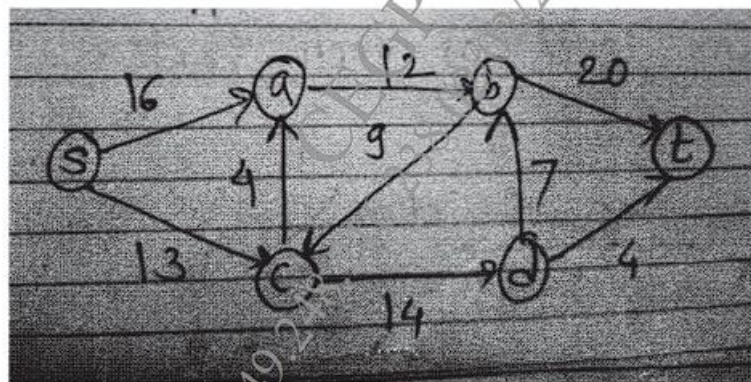
59. For the following graph find different cut sets and identify the max flow in the given network.



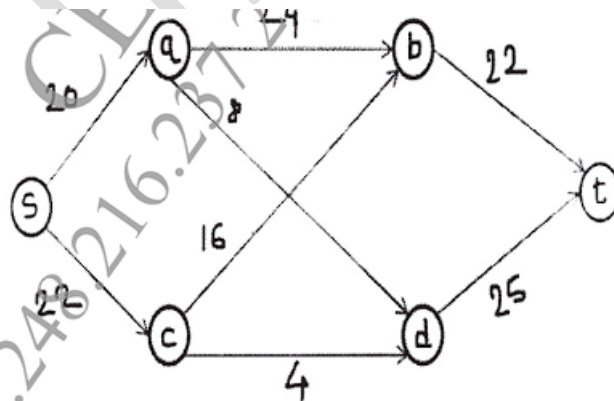
60. For the following transport network find the maximum flow using max flow min cut theorem.



61. Using the labelling procedure to find maximum flow in the transport network. Determine the corresponding minimum cut.

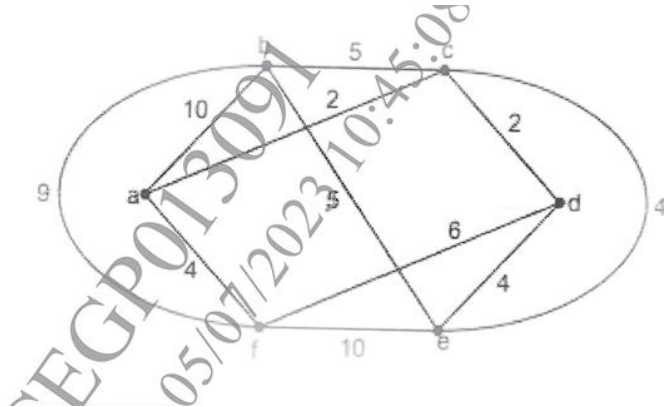


62. Determine the maximum flow in the transport network given below. Determine the corresponding min cut.

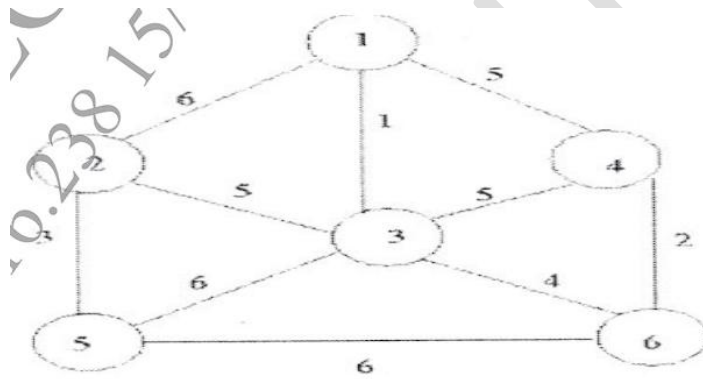


Minimum Spanning Tree — Prim's & Kruskal's

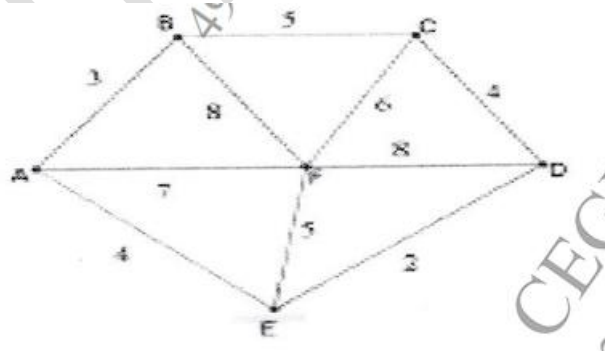
63. Use Kruskal's algorithm to find the minimum spanning tree for the connected weighted graph G.



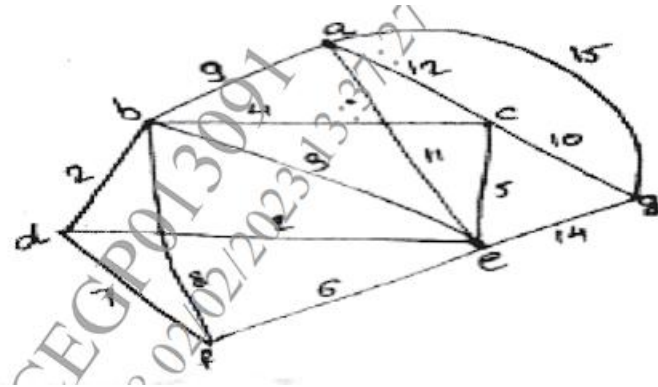
64. Construct Minimal spanning tree for the following graphs using Prim's algorithm.



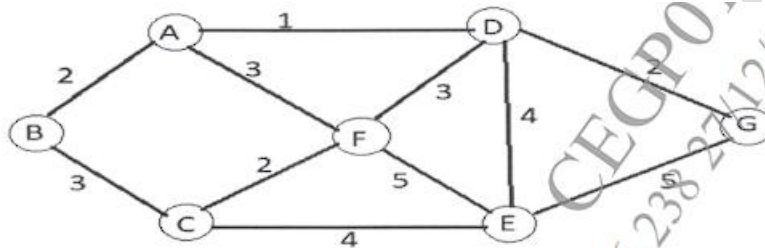
65. Construct Minimal spanning tree of the following graphs using Kruskal's algorithm.



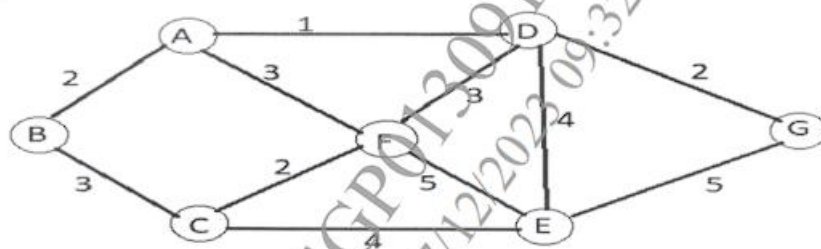
66. Give the stepwise construction of minimum spanning tree using Prim's algorithm for the following graph. Obtain the total cost of MST.



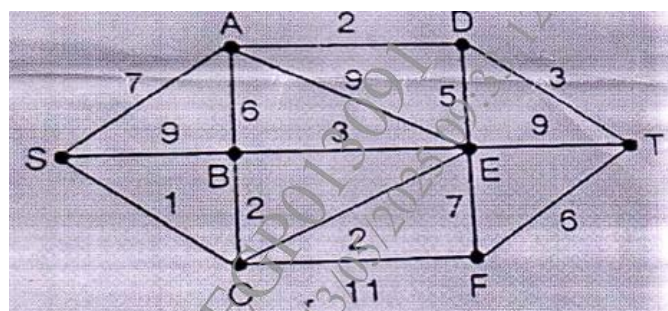
67. Find minimum Spanning tree using Prim's algorithm.



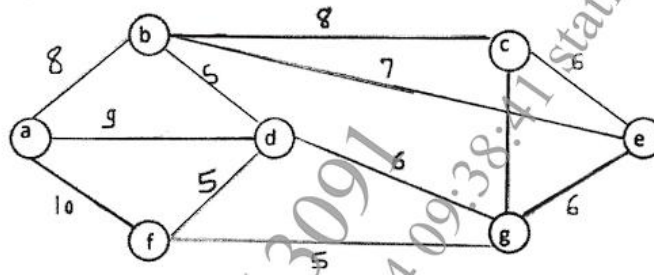
68. Find minimum spanning tree using Kruskal's Algorithm.



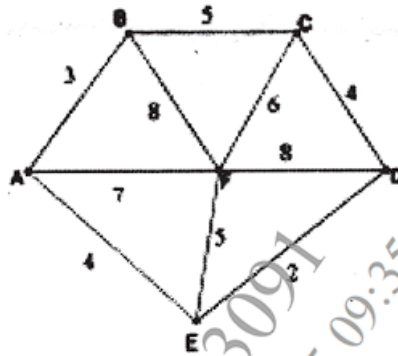
69. Use Kruskal's algorithm to find the minimum spanning tree for the connected weighted graph G.



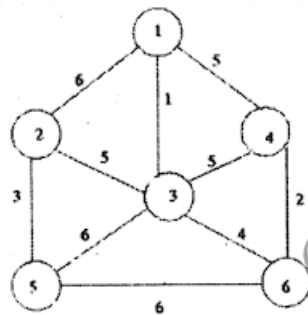
70. Use Prim's algorithm to construct a minimal spanning tree starting at vertex a.



71. Construct Minimal spanning tree for the following graphs using Prim's algorithm.



72. Construct Minimal spanning tree for the following graphs using Kruskal's algorithm.



UNIT IV — Introduction to Graph Theory

Graph Terminology & Types

73. Define the following terms with suitable example:

- i) Factor of graph
- ii) Weighted Graph
- iii) Bipartite graph

74. Define the following terms with example:

- i) Complete graph
- ii) Regular graph
- iii) Bipartite graph
- iv) Complete bipartite graph
- v) Paths and Circuits

75. Define the following terms with example:

- i) Complete graph
- ii) Regular graph
- iii) Bipartite graph
- iv) Complete bipartite graph
- v) Paths and Circuits
- vi) Acyclic Graph
- vii) Complement of a graph

76. Define the graph K_n and $K_{m,n}$.

77. Draw a complete bipartite graph on 2 and 4 vertices $K_{2,4}$ and 2 and 3 vertices $K_{2,3}$.

78. Under what condition $K_{m,n}$ will have Eulerian circuit.

79. Explain the following:

- i) Bipartite graph
- ii) Planar graph
- iii) Eulerian graph

80. Explain the following:

- i) Spanning tree
- ii) Colouring of graph
- iii) Bipartite graph

81. Explain the terms adjacency matrix and incidence matrix.

82. Explain with example:

- i) Bipartite Graph
- ii) Connected Graphs

83. Is it possible to draw a simple graph with 4 vertices and 7 edges? Justify.

84. Explain Edge connectivity and Vertex Connectivity with suitable example.

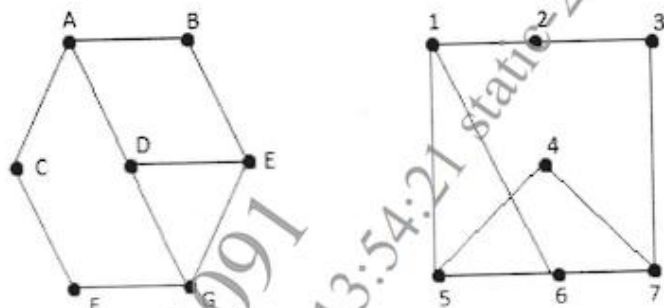
85. Is it possible to construct a graph with 12 nodes such that 2 of the nodes have degree 3 and the remaining have degree 4?

86. Explain the following statement with example: "Every graph with chromatic number 2 is a bipartite graph."

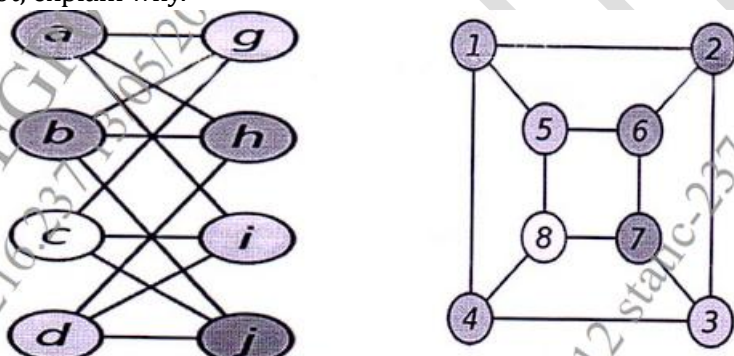
Graph Isomorphism

87. Draw all isomorphic graphs on vertices 2 and 3. Also draw all non-isomorphic graphs on 2, 3, and 4 vertices.

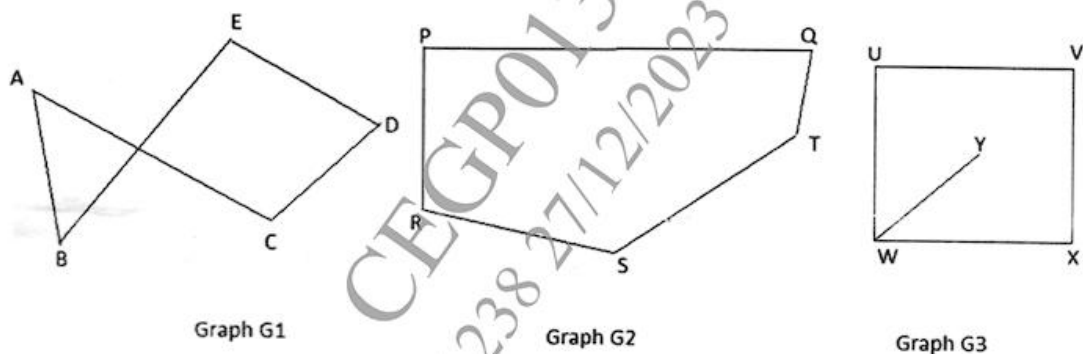
88. Show that the following graphs are isomorphic.



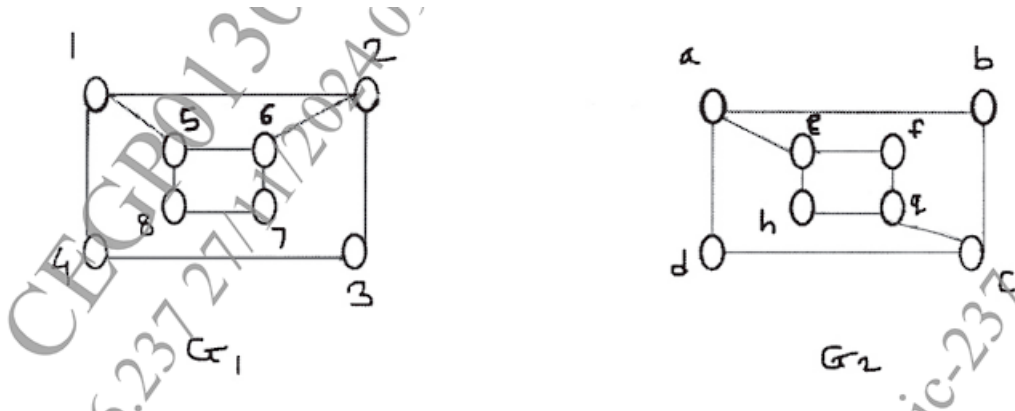
89. The graphs G and H with vertex sets $V(G)$ and $V(H)$ are drawn below. Determine whether or not G and H are isomorphic. If isomorphic, give a function $g: V(G) \rightarrow V(H)$ that defines the isomorphism. If not, explain why.



90. What is Graph isomorphism? Which of the following graphs are isomorphic? Justify your answer.



91. Whether the following pairs of graphs are isomorphic or not?



Euler & Hamilton Paths

92. List and explain the necessary and sufficient conditions for Hamiltonian and Eulerian path with suitable examples.

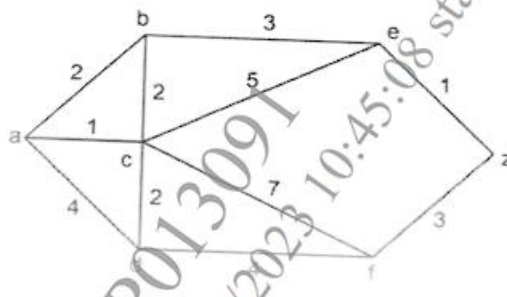
93. Determine which of the graphs below represents Eulerian circuit, Eulerian path, Hamiltonian circuit and Hamiltonian Path. Justify your answer. (Diagram in Nov-Dec 2022 paper)

94. Explain with suitable example:

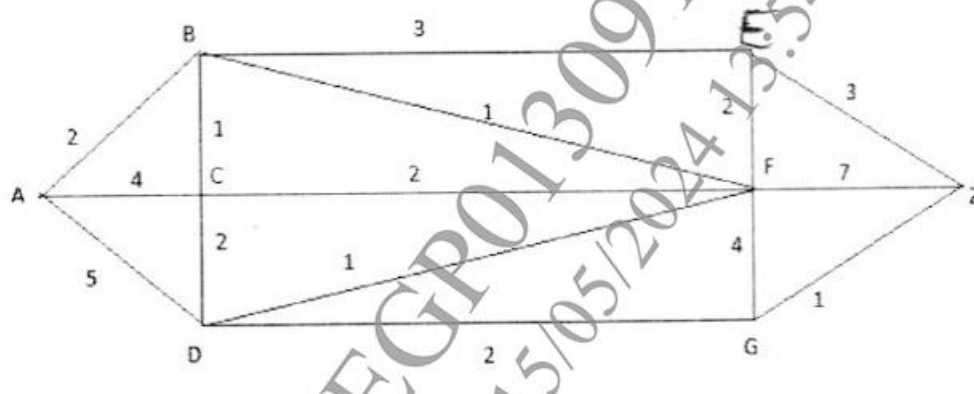
- i) Euler path & Euler circuit
- ii) Hamilton path & Hamilton circuit

Dijkstra's Shortest Path Algorithm

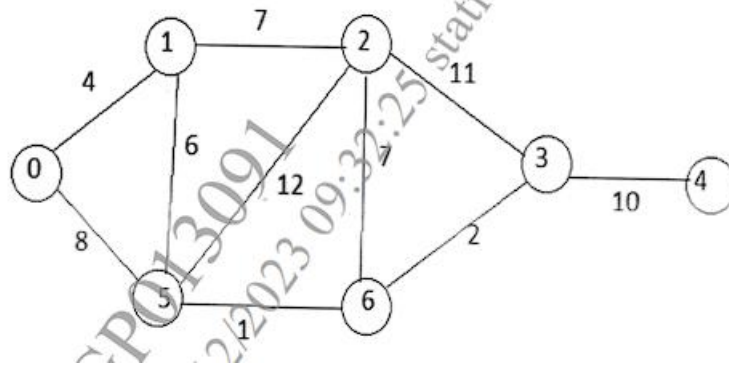
96. Find the shortest path between a-z for the given graph using Dijkstra's algorithm.



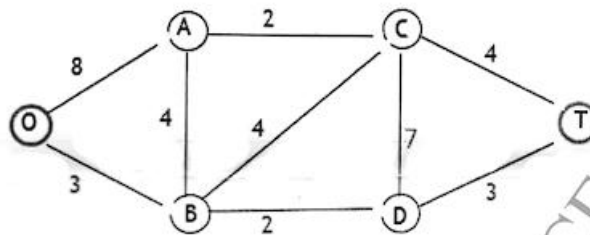
97. Use Dijkstra's algorithm to find the shortest path between A and Z.



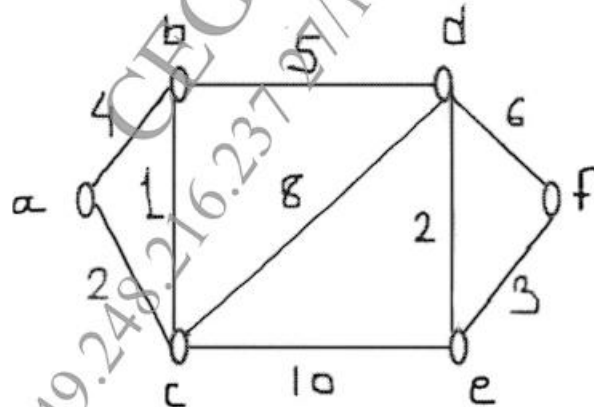
98. Find shortest path from vertex '0' to vertex '4' using Dijkstra's algorithm.



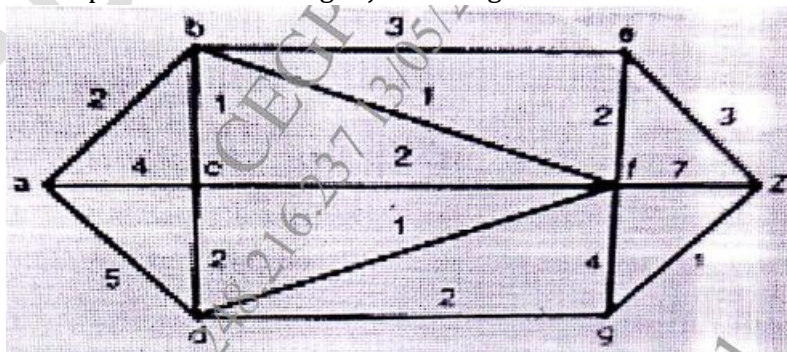
99. Find shortest path from vertex 'O' to vertex 'T' using Dijkstra's algorithm.



100. Use Dijkstra's algorithm to find the shortest path from vertex a to f.



101. Find the shortest path for a to z using Dijkstra's algorithm.



Planar Graphs

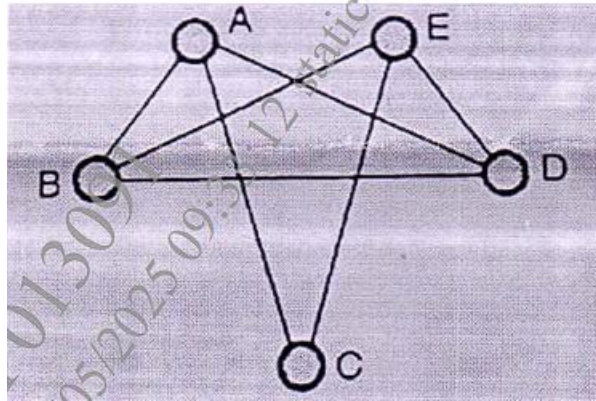
102. A connected planar graph has nine vertices with degree 2,2,2,3,3,3,4,4,5. Find:

- i) Number of edges
- ii) Number of faces
- iii) Construct two such graphs

103. What is a planar Graph? A simple planar graph G contains 20 vertices and degree of each vertex is 3. Determine the number of regions in planar graph G .

104. Prove that if G is a connected planar graph with N vertices, E edges and R regions then $N - E + R = 2$.

105. Convert the following non-planar graph into a planar graph. Also show the validity of Euler's formula for the given graph.



106. Show that in a connected planar graph with 6 vertices and 12 edges, each of the regions is bounded by 3 edges.

UNIT V — Counting Principles & Algebraic Structures
Permutations & Combinations

107. From a group of 7 men and 6 women, five persons are to be selected to form a committee so that at least 3 men are there. In how many ways can it be done?

108. Suppose repetitions are permitted:

- i) How many three-digit numbers can be formed from digits 2,3,4,5,7,9?
- ii) How many are multiples of 10?
- iii) How many are even?

109. Five pencils and 5 pens are to be arranged in a row. In how many ways if:

- i) All pencils must be arranged together
- ii) No two pencils should be kept together
- iii) One pen and one pencil must be arranged together

110. Find the number of permutations that can be made out of the letters of:

- i) MISSISSIPPI
- ii) ASSASSINATION

111. How many automobile license plates can be made if each plate contains two different letters followed by three different digits? Solve if the first digit cannot be zero.

112. How many 3-digit numbers can be formed from digits 2,3,5,6,7,9, which are divisible by 5 and none of the digits is repeated?

113. How many 6-digit odd numbers greater than 6,00,000 can be formed from digits 5,6,7,8,9,0:

- i) If repetition is allowed
- ii) If repetition is not allowed

114. In how many different ways can the letters of the word 'OPTICAL' be arranged so that the vowels always come together?

115. If a committee has eight members:

- i) How many ways can the committee members be seated in a row?
- ii) How many ways can the committee select a president, vice-president and secretary?

116. In a certain country, the car number plate is formed by 4 digits from 1–9 followed by 3 letters. How many number plates can be formed if neither digits nor letters are repeated?

117. The company has 10 members on its board of directors. In how many ways can they elect a president, a vice president, a secretary and treasurer?

118. Find the eighth term in the expansion of $(x+y)^{13}$.

119. A box contains 6 white and 5 black balls. Find the number of ways 4 balls can be drawn if:

- i) Two must be white
- ii) All of them must have the same colour

120. In how many ways can one distribute 10 apples among 4 children?

121. Use Binomial theorem to expand $(X^4 + 2)^3$.

122. How many bit strings of length 8 bits can be constructed which will either start with '1' or end with '00'?

123. In how many ways can 6 Boys and 2 Girls be seated in a row such that:

- i) 2 Girls are sitting together
- ii) 2 Girls are not sitting together

124. How many bit strings can be formed of length 10 bits which contain:

- i) At least four 1's
- ii) At most four 1's

125. How many bit strings of length 10 can be formed which will contain either 5 consecutive 0s or 5 consecutive 1s?

126. A zip code contains 6 digits. How many different zip codes can be made with digits 0–9 if:

- i) No digit is used more than once
- ii) The first digit is not '0'

127. Use the Binomial theorem to expand $(3a - 2b)^6$.

128. Suppose repetitions are not possible:

- i) How many three-digit numbers can be formed using digits 2,3,4,5,7,9?
- ii) How many of these numbers are less than 400? How many are even?

129. 12 persons are made to sit around a table. Find the number of ways they can sit such that 2 specific persons are not together.

130. A student has to answer 10 out of 13 questions in an exam:

- i) How many choices if he must answer the first or second question but not both?
- ii) How many choices if he must answer exactly three out of first five questions?

131. Suppose repetitions are not permitted. In how many ways can 4-digit numbers be formed from digits 1,2,3,5,7,8? How many such numbers are less than 4000?

132. Find the number of ways of arranging the letters of the word TENNESSEE all at a time. Find if the first two letters must be E.

133. A committee of 3 persons is to be formed from a group of 2 men and 3 women. In how many ways can this be done?

134. Find the number of arrangements that can be made out of the letters of:

- i) ASSASSINATION
- ii) GANESHPURI

135. What is the number of ways of choosing 4 cards from a pack of 52 playing cards? In how many of these:

- i) Are face cards?
- ii) Two are red cards and two are black cards?

136. Out of 4 officers & 10 clerks, a committee of 2 officers & 3 clerks is to be formed. In how many ways if:

- i) Any officer & clerk can be included
- ii) A particular clerk must be in committee
- iii) A particular officer cannot be in committee

137. How many words with or without meaning can be formed using the letters of the word EQUATION using each letter exactly once?

138. Salad is made with a combination of one or more eatables. How many different salads can be prepared from onion, carrot, cabbage & cucumber?

139. What is the coefficient of x^9 in the expansion of $(2-x)^{19}$?

140. A committee including 3 boys and 4 girls is to be formed from a group of 10 boys and 12 girls. How many different committees can be formed?

141. How many 4-letter words with or without meaning can be formed out of the letters of the word 'LOGARITHMS', if repetition of letters is not allowed?

142. A box contains 4 red, 3 white and 2 blue balls. Three balls are drawn at random. Find the number of ways of selecting balls of different colours.

Algebraic Structures

143. Define with examples:

- i) Groupoid
- ii) Semigroup
- iii) Monoid
- iv) Abelian group
- v) Subgroup

144. Define with examples:

- i) Properties of binary operation
- ii) Ring with unity
- iii) Fields
- iv) Integral Domain

145. Find the number of codes generated by the given check matrix H. Also find all code words.

1 1 0 1 0 0

0 1 1 0 1 0

1 0 1 0 0 1

146. Let $(A, *)$ be a monoid such that for every $x \in A$, $x * x = e$ where e is the identity element. Show that $(A, *)$ is an abelian group.

147. Define with examples:

- i) Cyclic group
- ii) Abelian group
- iii) Cosets

148. Let $Z_n = \{0, 1, 2, \dots, n-1\}$. Construct the multiplication table for $n = 6$. Is $(Z_n,)$ an abelian group where $ab =$ remainder of $a*b$ divided by n ?

149. Let $(A, *)$ be a group. Show that $(A,)$ is an abelian group iff $a^2 * b^2 = (ab)^2$.

150. Define with examples:

- i) Group codes
- ii) Subgroup
- iii) Integral domain

- 151.** Let $(A, *)$ be an algebraic system where $*$ is a binary operation such that for any $a, b \in A$, $ab = a$.
- i) Show that $*$ is an associative operation.
 - ii) Can $*$ ever be a commutative operation?
- 152.** Prove that the set Z of all integers with binary operation $*$ defined by $a*b = a+b+1$ for all $a, b \in Z$ is an abelian group.
- 153.** Let $R = \{0, 60, 120, 180, 240, 300\}$ and $*$ binary operation so that $a*b$ is the overall angular rotation corresponding to successive rotations by a and by b . Show that $(R, *)$ is a group.
- 154.** Following is the incomplete operation table of a 4-element group. Complete the last two rows.
- | | | | | |
|-----|---|---|---|---|
| $*$ | e | a | b | c |
| e | e | a | b | c |
| a | a | b | c | e |
| b | | | | |
| c | | | | |
- 155.** Explain Algebraic system and properties of binary operations.
- 156.** Consider the set Q of rational numbers and let $a*b = a+b-ab$:
- i) Find $3*4$
 - ii) $2*(-5)$
 - iii) $7*(1/2)$
- Is $(Q, *)$ a semigroup? Is it commutative?
- 157.** Let $Z_4 = \{0, 1, 2, 3\}$ and operation $+$ defined as: $a+b = a+b$ if $(a+b) < 4$; $a+b = a+b-4$ if $(a+b) \geq 4$. Determine if Algebraic System $(Z_4, +)$ is an abelian group or not.
- 158.** Explain:
- i) Integral domain
 - ii) Field
- 159.** Let $A = \{0, 1, 2, 3\}$ and operation $\odot$ defined as $a \odot b = a.b \% 4$. Determine if algebraic system $(A, \odot)$ is monoid or not.
- 160.** Let $Z_n = \{0, 1, \dots, n-1\}$. Consider operation $+$ defined as "addition Modulo 5" and $*$ as "multiplication modulo 5". Does the algebraic system $(Z_5, +, *)$ form a Ring?
- 161.** Explain the following properties of Algebraic structure with example:
- i) Identity
 - ii) Inverse
- 162.** Explain the following terms giving suitable example:
- i) Monoid
 - ii) Group
 - iii) Cyclic group
- 163.** Let $(Q, *)$ be an algebraic system where $ab = a+b-ab$ for every $a, b \in Q$. Determine whether $(Q, *)$ is a group.

164. Show that $S = \{a+b\sqrt{2} ; a, b \in \mathbb{Z}\}$ for operations $+$ and $*$ is an integral domain.

165. Explain the following terms giving suitable example:

- i) Homomorphism of groups
- ii) Integral domain
- iii) Abelian group

166. Let $Z_n = \{0, 1, 2, \dots, n-1\}$. Let $*$ be a binary operation such that $a*b = \text{remainder of } ab \text{ divided by } n$.

- i) Construct the table for $n = 4$
- ii) Show that $(\mathbb{Z}_n, *)$ is a semigroup for any n .

167. What is Hamming distance? Find minimum distance generated by parity matrix $H = 110$
 011 101 100 010 001 . How many errors can it detect or correct?

168. Define with examples:

- i) Ring with unity
- ii) Fields
- iii) Integral Domain

169. Define:

- i) Semi-group
- ii) Field
- iii) Monoid

170. Define:

- i) Ring
- ii) Ring Homomorphism
- iii) Integral domain

171. Define with examples:

- i) Groupoid
- ii) Monoid
- iii) Abelian group

172. Prove that $(I, +)$ is an abelian group, where I is a set of all integers with respect to binary operation '+'.
